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Data-driven and privacy-preserving risk assessment method based on federated learning for smart grids



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Timely and precise security risk evaluation is essential for optimal operational planning, threat detection, and the reliable operation of smart grid. The smart grid can integrate extensive high-dimensional operational data. However, conventional risk assessment techniques often struggle with managing such data volumes. Moreover, many methods use centralized evaluation, potentially neglecting privacy issues. Additionally, Power grid operators are often reluctant to share sensitive risk-related data due to privacy concerns. Here we introduce a data-driven and privacy-preserving risk assessment method that safeguards Power grid operators' data privacy by integrating deep learning and secure encryption in a federated learning framework. The method involves: (1) developing a two-tier risk indicator system and an expanded dataset; (2) using a deep convolutional neural network-based model to analyze the relationship between system variables and risk levels; and (3) creating a secure, federated risk assessment protocol with homomorphic encryption to protect model parameters during training. Experiments on IEEE 14-bus and IEEE 118-bus systems show that our approach ensures high assessment accuracy and data privacy.

As the fundamental infrastructure of modern society, the power system plays a vital role in ensuring essential energy supply. Recent advancements in communication systems have brought about important changes, offering increased opportunities for controlling and monitoring the entire power grid. This technological progress has not only transformed the operational aspects of the power system but has also opened up avenues for energy management¹. The emergence of smart grids (SG), facilitated by advanced data information and communication technology (ICT), enhances the optimization of generation, storage, transmission, distribution, and utilization of green energy in an intelligent and efficient manner. This evolution greatly boosts the operational efficiency of the power grid².

However, the advancement of intelligence in systems is accompanied by a rise in system complexity. The frequent interplay between information systems and physical systems blurs security boundaries, potentially introducing security threats through close interactions. The repercussions of security threats within the information system domain extend throughout the entire energy data lifecycle, impacting not only the information system itself but also potentially affecting the closely interconnected physical system³. Cyber-attacks and system failures are no longer confined to virtual spaces, and they can swiftly propagate to the physical realm, resulting in damage to actual equipment and power instability, posing a severe threat to power supply stability. Incidents such as the power outages experienced by

the Ukraine and Israel power grids have underscored the ability of attackers to exploit cyber vulnerabilities to harm critical infrastructure, emphasizing the considerable risk posed to the reliable operation of power systems^{4,5}. Moreover, the inherent stochastic nature of power systems, characterized by volatility and unpredictability in power demand, coupled with uncertainties in grid operations stemming from factors such as natural disasters, equipment malfunctions, and human errors, further contribute to a complex risk environment for smart grid (SG) operations. Neglecting these risks could jeopardize the stable operation of SG and potentially lead to serious accidents. Therefore, conducting a comprehensive and prompt assessment of security risks and implementing effective management strategies is crucial for safeguarding the stable operation of SG.

The SG can collect a substantial amount of data on high peacekeeping and multiple types of data by integrating advanced metering infrastructure, control technology, and communications technology⁶. Traditional risk assessment methods can be divided into analytical methods and simulation methods. The former portrays the structure and internal logical relationships of the system based on strict mathematical models, solving the models accurately through recursion or iteration. The latter, such as the Monte Carlo Simulation (MCS), utilizes random numbers to characterize the probability distribution of the operating state of each component of the power system, and then estimates the risk indicators of the power system

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through simulation. Both methods rely heavily on accurate, mechanism-driven models of the power system, require a substantial amount of computational resources⁷, and do not have sufficient capacity to handle the large amount of data introduced by SG. Benefiting from the advancements in artificial intelligence technology, data-driven approaches that incorporate machine learning (ML) have revitalized risk assessment. These approaches leverage large-scale data collected by SG to enhance the performance of SG. Despite the offline training procedures can be time-consuming, data-driven approaches can provide immediate responses in online contexts, improving the speed of risk assessment runs. However, the aforementioned methods employ a centralized training approach that requires all data to be transferred to a centralized location with powerful computing and storage capabilities. This operational mode is susceptible to the threat of data leakage. More importantly, SG's operational data and control information are highly sensitive. Each individual system data contains a certain number of risk samples, often reflecting the system's security weaknesses. Once attackers gain access to critical data, they can better plan attack strategies, execute a more targeted malicious data attack, or implement a physical attack by accessing information about critical devices.

Considering the aforementioned challenges, this paper introduces a Federated Learning⁸-based Risk Assessment (FedRA) method to facilitate distributed risk assessment. By leveraging federated learning (FL), decentralized model training is implemented across various power system nodes, eliminating the need to share original data to uphold privacy. To quantify the risks encountered by the power system, a comprehensive power system

security risk assessment index system is initially developed. Subsequently, a risk assessment model based on Convolutional Neural Networks (CNN) is established to detect risk patterns within the power system, capitalizing on its effective feature extraction capabilities. The CNN-based risk assessment model is then trained using a federated learning approach to ensure efficient risk assessment while safeguarding data privacy. The core idea of this paper is illustrated in Fig. 1.

In summary, our contribution can be summarized as follows.

- A two-tier risk assessment index system is constructed, in which risk indicators such as node voltage overrun, line power overload and system loss of load are incorporated into the first-tier risk assessment index system, and the three are weighted to obtain the second-tier comprehensive risk indicators. An extended dataset is generated based on the above indicator system to provide data support for training the risk assessment model.
- A CNN-based risk assessment model is designed which employs a data-driven approach for security risk identification to improve the accuracy and real-time performance of the risk assessment task.
- Combining FL and homomorphic encryption proposes a scheme called FedRA for risk assessment, which allows each PGO node to train collaboratively using local datasets while preserving the privacy of sensitive data.
- Extensive experiments are carried out on the IEEE 14-bus system and the IEEE 118-bus system, and the experimental results show that our

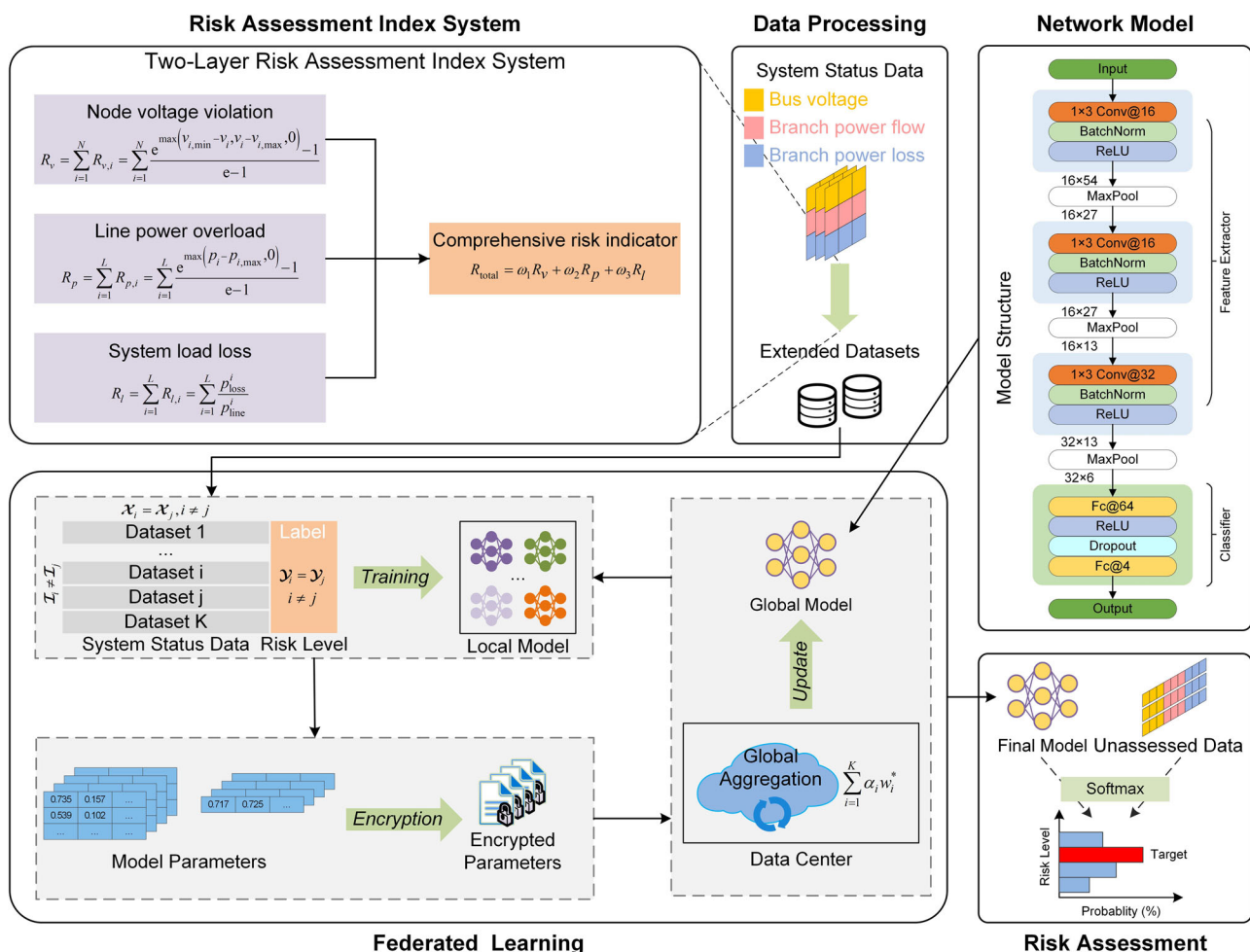


Fig. 1 | Framework of FedRA. The risk index system is employed in the construction of the dataset, with participants utilizing the local dataset to collaboratively train a global model for the risk assessment task based on federated learning. $\mathcal{X}$ is the feature

space, $\mathcal{Y}$ is the label space, and $\mathcal{I}$ is the ID space for each dataset. α_i is the proportion of each node's dataset to the total, w_i^* is the local model parameter, and $\sum_{i=1}^K \alpha_i w_i^*$ is the aggregated ground model parameter.

FedRA scheme has a high evaluation accuracy while safeguarding node privacy.

Methods

Related work

Additional risk assessment methods. Risk assessment forms the foundation for ensuring the secure and stable operation of the power system. Most risk assessment methods are mainly implemented through analysis-based, simulation-based and ML-based methods.

The analytical methods involve using mathematical and statistical models to analyze and quantify risks in the power system. These methods are based on mathematical modeling of the physical characteristics of the power system⁹, and quantifies the risk by calculating the probability of different events occurring and the losses caused by them. Xu et al.¹⁰ proposed a hybrid multi-surrogate (HMS) method based on polynomial chaos expansion, which reevaluates low probability rare events through probabilistic analysis. The HMS method enhances the computational efficiency of rare event risk assessment while preserving the required accuracy. To accurately deal with multiple types of random variables, Dong et al.¹¹ proposed a non-intrusive probabilistic multi-energy flow calculation method, which calculates various system risk indices using probability density functions and quantifies the risk level using the conditional value-at-risk method.

The simulation method is founded on probability theory and mathematical statistics. It simulates the complex interaction and dynamic behavior of the power system in detail, to obtain the probabilistic statistical characteristics of various random variables. Chen et al.¹² proposed a risk assessment method based on a hybrid meteorological-electric system simulation platform with an adaptive step alternation simulation strategy to model the alternate execution of each process. MCS is a widely used simulation method due to its adaptability to system size and complexity, as well as its convergence and computational accuracy^{13,14}. However, the computational time consumption greatly increases due to the large number of sample tests required to achieve high accuracy. Huang et al.¹⁵ proposed an efficient importance sampling technique that accelerates the MCS process by reducing variance and improving the estimation efficiency of low probability violations.

Due to the high efficiency of ML in handling massive high-dimensional data, it has been widely applied in risk assessment research. Ren et al.¹⁶ proposed a fully data-driven method using a generative adversarial network to process incomplete Phasor Measurement Unit data for pre-fault dynamic security assessment of the grid. Leite et al.¹⁷ proposed a model that combines plain Bayes and Jenks natural break classifiers to calculate the probability of a grid risk event occurring and determine its impact based on a time-series formula. To effectively address potential faults in the power system, Ren et al.¹⁸ utilized transfer learning to train assessment models that cover a wide range of fault scenarios, which improves the generalization and efficiency of security assessment. Wu et al.¹⁹ proposed a risk level classification model for each risk event based on fuzzy C-mean clustering and Mamdani fuzzy inference method to determine the severity of risk events. Bugaje et al.²⁰ proposed an assessment method based on decision tree classification, which improves the accuracy and real-time performance of power system security assessment by optimizing multiple objectives such as interpretability, cost and accuracy.

Comments: Both analytical and simulation methods are based on mechanistic models of the system, and accurate mathematical modeling is complex for real SG environments. Additionally, the computational cost of dealing with the huge amount of data collected by SG is still high, even when the variance of the simulation is reduced²¹. The inputs and outputs of the above models are still explicitly variable (i.e., input system state data and output system risk value or level), which motivates the use of ML methods in risk assessment. However, the aforementioned ML methods still employ centralized data processing and computation, which may raise issues of data privacy and security. Furthermore, relying solely on a single data source may

not cover all potential risk scenarios, leading to the inability to accurately identify rare risk types.

Federated learning in SG. As mentioned in the previous section, centralized model training requires the aggregation of all data into a central node for processing. However, system data containing risk samples is often highly sensitive, and individual system operators typically have only limited risk samples that reflect unique data characteristics of their systems. Considering privacy and security concerns, data owners are generally reluctant to share their data.

As an emerging distributed machine learning method, federated learning has been successfully implemented in the field of SG. For instance, Wang et al.²² proposed a distributed FL-based feature recognition method for electricity users, which can protect the privacy of each electricity retailer without sharing the original smart meter data. In response to the privacy issues related to centralized detection methods, Li et al.²³ combined the Paillier cryptosystem with FL to design the SecFed scheme, and constructed a Transformer-based false data injection attack detection (FDIA) model to detect covert FDIA in the SG. Badr et al.²⁴ introduced FL to construct a global energy prediction model for net metering systems that accurately predict future energy demand while maintaining privacy and efficient communication. Jithish et al.²⁵ proposed an anomaly detection method for SG based on FL, which allow training machine learning models on smart meters without sharing data with a centralized server, thereby protecting user privacy.

Comments: The schemes mentioned above in the literature all use FL methods to train models on distributed data samples without sharing the original data to protect privacy. However, it is still possible for attackers to infer the original data information from the local training parameters²⁶, so encryption of the model parameters is particularly necessary. In addition, the application of FL in security risk assessment in smart grids has not been fully explored. The privacy and security issues involved in the assessment process also deserve further consideration. To this end, this paper proposes a FL-based scheme for security risk assessment, which will be detailed in the subsequent sections.

Federated learning

FL serves as one of the representatives of privacy-preserving machine learning that retains the basic architecture of distributed learning, which involves localized data processing on multiple nodes but excels in privacy protection. A few core features of FL are as follows:

1. Private data: Unlike distributed learning, where data is stored centrally and then distributed, FL keeps each participant's raw data locally instead of uploading it to a central server.
2. Secure communication: Only the model parameters should be transmitted, not the raw data. Perturbation and encryption methods are commonly employed to handle model parameters, ensuring data privacy during transmission.
3. Collaborative learning: Coordinating numerous remote devices to collectively train a global model, each participant has its own independent dataset with a distribution that may differ from the global one.
4. Approach ideal performance: All participants obtain a collectively better model than training individually. The model's final performance approaches that of an ideal model, which is achievable when all data are available for centralized training.

FL has been widely applied across multiple machine learning scenarios within the domain of SG. For better explanation, we will describe the FL in the context of risk assessment. Let $\mathcal{P} : \{P_1, P_2, \dots, P_K\}$ denote the set of K power grid operator nodes participating in FL, where each node possesses a private system risk dataset $\{D_i\}_{i=1}^K$. Each row of D_i represents a data sample, and each column denotes specific data features. Then, let $\mathcal{X}$ be the feature space, $\mathcal{Y}$ be the label space, and $\mathcal{I}$ for the ID space of each data sample. Based on the different distributions of dataset in the feature space and ID space

among different users, FL can be divided into two common scenarios: horizontal FL and vertical FL. In this paper, we focus on risk assessment under horizontal FL, where the data distribution among different participants can be described as

$$\mathcal{I}_i \neq \mathcal{I}_j, \mathcal{X}_i = \mathcal{X}_j, \mathcal{Y}_i = \mathcal{Y}_j \quad \forall D_i, D_j, i \neq j \quad (1)$$

where i, j denote two independent PGOs that manage power node systems of similar size under different operational conditions. Each piece of data is sampled from a single runtime of the power system (different sample space), the size of $\mathcal{X}$ is related to the scale of the system which corresponds to multiple critical system state parameters (same feature space), and $\mathcal{Y}$ represents the system risk level at a particular runtime. The risk assessment problem is viewed as a classification problem. Each participant trains a local model $\{w_i\}_{i=1}^K$ using their own data, which is then aggregated by a central server to collaboratively build a global classification model w_G .

Homomorphic encryption

Homomorphic encryption is a cryptographic algorithm of encryption based on mathematical computation. Data processed by homomorphic encryption can perform the same operations as the original unencrypted data and obtains the same result after decryption. We introduce the Cheon-Kim-Kim-Song (CKKS)²⁷ fully homomorphic encryption algorithm to protect the model parameters, which supports both additive homomorphic operations and multiplicative homomorphic operations on float numbers and vectors. The components of the algorithm are as follows:

1. $\text{KeyGen}(1^\lambda) \rightarrow (\text{PK}, \text{SK}, \text{evk})$: The algorithm takes the security parameter λ as input and outputs the public key PK for encryption, the private key SK for decryption, and the auxiliary key evk for assisting the homomorphic multiplication operation.
2. $\text{Encrypt}(\text{PK}, m) \rightarrow c$: For a given plaintext m , the algorithm encrypts plaintext m using public key PK and outputs ciphertext c . To make the ciphertext/plaintext boundary clearer, we use $\llbracket m \rrbracket$ to denote the ciphertext space corresponding to plaintext A .
3. $\text{Add}(c_1, c_2) \rightarrow c_3$: The homomorphic addition algorithm takes the ciphertexts of m_1 and m_2 as input and outputs the ciphertext c_3 of $m_1 + m_2$, i.e., $\llbracket m_1 \rrbracket \oplus \llbracket m_2 \rrbracket = \llbracket m_1 + m_2 \rrbracket$, and $\oplus$ denotes the addition on the ciphertext.
4. $\text{Mult}(c_1, c_2, \text{evk}) \rightarrow c_4$: The homomorphic multiplication algorithm takes the ciphertexts of m_1 and m_2 as input and outputs the ciphertext c_4 of $m_1 \times m_2$ using an auxiliary key evk , i.e., $\llbracket m_1 \rrbracket \otimes \llbracket m_2 \rrbracket = \llbracket m_1 \times m_2 \rrbracket$, where $\otimes$ denotes the addition on the ciphertext.
5. $\text{Decrypt}(\text{SK}, c) \rightarrow m$: For a given ciphertext c , the algorithm decrypts the ciphertext c using the private key SK and outputs the plaintext m .

Risk indicator system

As mentioned earlier, there are multiple security risk factors in SG, the deep coupling of information physics makes it extremely easy for the impacts caused by malicious cyber-attacks to propagate to the physical system, and the uncertainty and randomness of the power system itself makes the system even more vulnerable. Considering that the most direct impact of these risk factors on the stable operation of the SG is the occurrence of overload in transmission lines or load shedding events due to errors in certain power load nodes, we mainly incorporate the risk indicators of node voltage violation, line power overload and system load loss into the first tier of the risk assessment index system. By weighting these indicators, we obtain a second-layer comprehensive risk indicator.

Node voltage violation. Voltage stability is a crucial indicator that measures whether a power system can maintain an appropriate voltage level under varying load conditions. Poor voltage stability may lead to instability in part or the entirety of the power system. Lower voltage levels increase the energy consumption of the system operation. Conversely, excessive voltage levels may lead to overload and damage to equipment

within the SG. Voltage violation indicator is measured by comparing real-time monitored or predicted node voltages with corresponding voltage safety operating limits, which can be expressed as

$$R_v = \sum_{i=1}^N R_{v,i} = \sum_{i=1}^N \frac{e^{\max(v_{i,\min} - v_i, v_i - v_{i,\max}, 0)} - 1}{e - 1} \quad (2)$$

where N is the total number of system nodes, i is the node number, $R_{v,i}$ represents the voltage violation risk value at i -th node, v_i represents the real-time monitored voltage level at i -th node, and $v_{i,\min}$ and $v_{i,\max}$ are the lower and upper limits of the voltage amplitude at i -th node, respectively, during normal operation of the system.

Line power overload. The line power overload risk indicator focuses on whether the power of an operating electric transmission line exceeds its rated value. Overloading may occur when the line's current exceeds its design capacity. Prolonged overload conditions can lead to heating, insulation degradation, and even breakage of the line. The line power overload indicator is calculated by comparing the deviation between the actual power and the safe line power limit using the following formula:

$$R_p = \sum_{i=1}^L R_{p,i} = \sum_{i=1}^L \frac{e^{\max(p_i - p_{i,\max}, 0)} - 1}{e - 1} \quad (3)$$

where L is the total number of system branches, i is the branch number, $R_{p,i}$ represents the risk value of power overload of i -th branch, p_i is the real-time monitored power level of i -th branch, and $p_{i,\max}$ is the safe upper limit of power transmitted by the branch, which is set to 1.3 times the rated power value²⁸.

System load loss. The system load loss risk indicator measures the extent to which the power system loses supply load for various reasons, and is calculated based on the value of load loss transmitted by the system lines and the ideal amount of load transmitted by the line loads. The formula is as follows:

$$R_l = \sum_{i=1}^L R_{l,i} = \sum_{i=1}^L \frac{p_{\text{loss}}^i}{p_{\text{line}}^i} \quad (4)$$

where L is the total number of system branches, i is the branch number, $R_{l,i}$ is the load loss severity on i -th branch, which is equal to the ratio of the active power loss value p_{loss}^i on i -th branch to the actual active power flow value p_{line}^i .

Comprehensive risk indicator. In view of the impact of the first tier of risk assessment indicators on the SG, a comprehensive operational risk assessment indicator is introduced as the second tier of risk assessment indicators. By assigning certain weights to the above three risk indicators, the comprehensive risk index is calculated as follows:

$$R_{\text{total}} = \omega_1 R_v + \omega_2 R_p + \omega_3 R_l \quad (5)$$

where ω_1 , ω_2 , and ω_3 are the weights of node voltage violation, line power overload, and system load loss, respectively. We set the weights all to the same value to indicate equal importance among the three.

Building upon the above risk indicator system, specific risk values for each risk indicator can be calculated by sampling data from the power system under different operating conditions. Thus, the comprehensive security risk can be determined and the security risk level of the system can be classified. We classify risk levels into four categories: [Safe, Low Risk, Medium Risk, High Risk].

Risk assessment model based on CNN

In this section, we propose a data-driven model based on CNN to assess the risk level of SG. The risk assessment problem can be formulated as a four-classification problem, where the inputs to the CNN are the active power

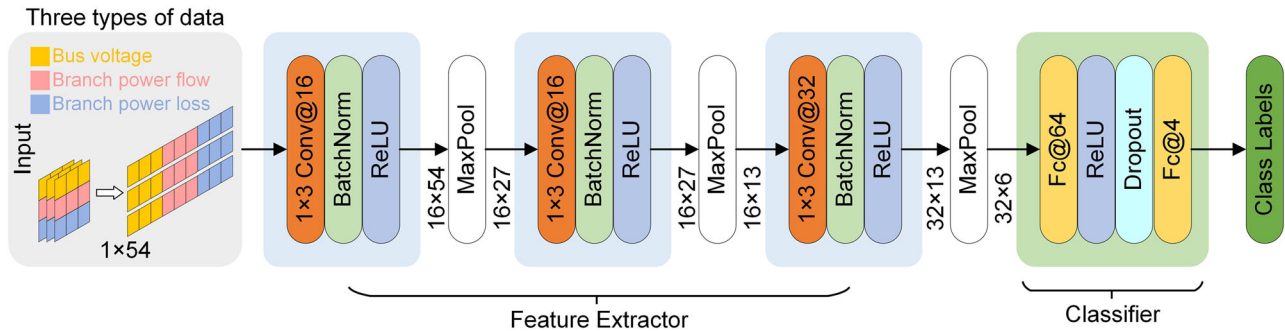
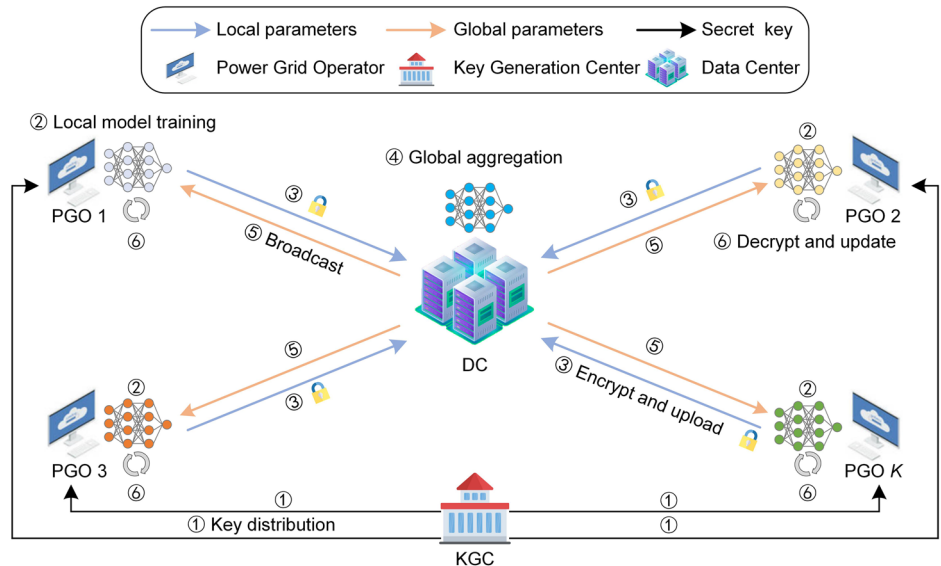


Fig. 2 | Basic structure of proposed model for risk assessment. The model consists of a feature extractor and a classifier that takes state data as input and outputs a risk level. 1×3 Conv@16: convolutional layer with 1×3 convolution kernel size and 16 channels. ReLU: activation function. Fc: full connection layer.

Fig. 3 | FedRA detailed process. The whole process can be divided into six fundamental steps, with the data flow represented by a series of colored lines. The accompanying illustration provides a detailed explanation of the meaning behind each icon. PGO: power grid operator, KGC: key generation center, DC: data center.



injection and voltage amplitude levels at the buses and the power loss at the branches, while the output is a certain risk level of the power system. The power operators utilize their local dataset $\{D_i\}_{i=1}^K$ to train a classification model $y = f(\omega_i, D_i)$, where ω_i denotes the model parameters of local model. The label y takes values from the set $\{0, 1, 2, 3\}$, corresponding to the four risk levels [Safe, Low Risk, Medium Risk, High Risk]. Specific details regarding the construction of the dataset required for model training will be extensively covered in the experiment section of this paper. We use CNN to analyze the deep structure of highly correlated data, thereby identifying anomalous characteristics of high-risk measurement samples compared to normal historical measurement samples. In contrast to traditional risk assessment methods, our approach is completely data-driven: no modeling of the power system is required, and the entire risk assessment process doesn't involve system mechanistic models and parameters. Instead, it directly assesses the risk level of the current system state by sampling the state data of the SG under different operating conditions.

Figure 2 shows details of the CNN-based model used for risk assessment, which consists mainly of an input layer, several feature extractors, and a classifier. Specifically, the input layer has a 1-dimensional input vector of length n , representing the system's state measurements at a single runtime. In the case of the IEEE 14-bus system, the input is a 1×54 (number of buses N : 14 + number of branches $L \times 2$: 20×2) 1-dimensional vector. The feature extractor consists of convolutional layer, batch normalization layer, activation function layer and maximum pooling layer, which uses convolutional computation to explore the potential patterns of the grid state data, and then extracts the most relevant information from the features obtained from the convolutional layer through pooling operations to reduce the dimensionality

of the data. Finally, the output of the pooling layer is flattened (i.e., concatenating features from multiple windows into a 1-dimensional output) and fed into the linear layer of the classifier, which outputs the final classification result via the sigmoid function. This model serves as the base model used in the FL framework proposed below, with fewer layers and parameters, which means that the model is more lightweight, contributing to reducing communication overhead and ensuring efficient collaboration in FL.

FedRA scheme

In the context of risk assessment using machine learning methods, a substantial amount of data is typically collected for model training. However, in risk assessment scenarios, multiple grid operators usually have only a limited number of risk samples. These data samples point to vulnerabilities in the power system, and operators may be reluctant to share them with other participants, so there is necessary to minimize the data exchange between different power nodes to protect their privacy. The key idea of the FedRA scheme is that multiple power nodes collaboratively training a risk assessment model in a distributed manner, with the sensitive data itself is not shared. The detailed process of this scheme is given in Fig. 3, which consists of four main stages: Initialization, Training and Encryption, Aggregation, and Decryption and Update.

(1) Stage 1 (Initialization): The aim of this stage is to establish a secure and reliable FL environment, completing the initialization process to ensure the stable operation of FedRA.

(a) Key Generation at KGC: Based on the homomorphic encryption system, the KGC is responsible for generating the public and private key pair (PK, SK) for the homomorphic encryption computations, and then sending

the key pair (PK,SK) to each PGO node for encrypting and decrypting the model weights.

(b) Global Model Initialization at DC: Before the entire collaborative training process starts, based on the risk assessment model described in the previous section, the DC generates an initial global model w_{global}^0 using a preset random seed. This initial global model is then transmitted to each PGO node for the first iteration of local training. This step ensures that all nodes have a common initial model when starting the training.

Algorithm 1. FedRA

Input: Initial global parameters w_{global}^0 , maximum communication rounds C , number of local epochs I , local datasets D_k , loss function L , key pair (PK,SK), $\mathcal{P} : \{P_1, P_2, \dots, P_K\}$

Output: Final model w_{global}

```

1 Initialize:  $\eta, \rho_1, \rho_2, \varepsilon, \alpha_k, w_{k,c}^0 = w_{\text{global}}^0$ ;
2 for  $c \leftarrow 1$  to  $C$  do
3   for  $P_k \in \mathcal{P}$  do
4     for  $i \leftarrow 1$  to  $I$  do
5       gradient calculation:
6        $g_i \leftarrow \nabla w_{k,c}^i L$ ;
7       update biased estimate:
8        $z_i \leftarrow \rho_1 z_{i-1} + (1 - \rho_1) g_i$ ;
9        $s_i \leftarrow \rho_2 s_{i-1} + (1 - \rho_2) g_i^2$ ;
10      calculate bias-corrected estimation:
11       $\hat{z}_i \leftarrow \frac{z_i}{(1 - \rho_1^i)}, \hat{s}_i \leftarrow \frac{s_i}{(1 - \rho_2^i)}$ ;
12      local parameters update:
13       $w_{k,c}^i \leftarrow w_{k,c}^{i-1} - \eta \frac{\hat{z}_i}{\sqrt{\hat{s}_i + \varepsilon}}$ ;
14      complete local iteration to generate  $w_{k,c}^*$ ;
15      encrypt  $w_{k,c}^*$  with public key PK by  $[[w_{k,c}^*]] \leftarrow \text{Encryption}(PK, w_{k,c}^*)$ ;
16      send ciphertext  $[[w_{k,c}^*]]$  to the DC;
17  DC aggregates the model parameters by  $[[w_{\text{global}}^c]] \leftarrow \text{Aggregation}(\alpha_k, w_{k,c}^*) = \sum_{k=1}^K \alpha_k [[w_{k,c}^*]]$ ;
18  broadcast global model to each PGO node;
19  for  $P_k \in \mathcal{P}$  do
20    decrypt  $[[w_{\text{global}}^c]]$  with private key SK by  $w_{\text{global}}^c \leftarrow \text{Decryption}(SK, [[w_{\text{global}}^c]])$ ;
21    update the local model;
22 return  $w_{\text{global}}$ ;

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(2) Stage 2 (Training and Encryption): In this stage, all PGO nodes complete the local model training and upload the encrypted model parameters, while retaining the training data locally. The specific workflow is shown in Algorithm 1.

(a) Local Training at PGO: For each node P_k participating in FL, the model parameter $w_{k,c}^i$ is obtained by training with its private dataset D_k during the local training process. we denote $w_{k,c}^*$ as the set of all weights and biases of the model, then the local risk assessment model can be expressed as:

$$y_k^i = f(w_{k,c}^i, D_i) \quad (6)$$

where i denotes the i -th iteration of local training, k denotes the k -th PGO node, and c is the c -th communication round for uploading the model. We use the Adam optimization algorithm to update the model parameters, and finally obtain the model parameters $w_{k,c}^*$ for the c -th communication iteration round.

(b) Model Encryption at PGO: After all PGO nodes have completed training, the local model parameters need to be uploaded to the DC for aggregation.

Unlike the direct uploading of model parameters, we use the CKKS homomorphic encryption algorithm to encrypt the local model, whose

support for float homomorphic operations aligns well with the characteristic that CNN model parameters are mostly signed float numbers. Therefore, in this stage, each node executes the following equation to encrypt the local model parameter $w_{k,c}^*$ using the public key PK.

$$\text{Encryption}(PK, w_{k,c}^*) \rightarrow [[w_{k,c}^*]] \quad (7)$$

where PK is the public key held by PGO, used for local encryption of model parameters, and $[[w_{k,c}^*]]$ denotes the ciphertext.

(3) Stage 3 (Aggregation): This stage is dominated by the DC, aiming to aggregate model parameters to generate the global model.

Global Aggregation at DC: Once the DC receives the local cryptographic models sent by each PGO node, the following secure aggregation algorithm is used to integrate each local model to obtain global model $[[w_{\text{global}}^c]]$, which is then transmitted to each local node.

$$\text{Aggregation}(\alpha_k, w_{k,c}^*) = \sum_{k=1}^K \alpha_k [[w_{k,c}^*]] \rightarrow [[w_{\text{global}}^c]] \quad (8)$$

where α_k denotes the proportion of the total dataset that each node contributes during training. This process operates in the ciphertext space, ensuring that the performance of the global model integrates the contributions of each node while protecting the privacy of the node.

(4) Stage 4 (Decryption and Update): In this stage, the local nodes decrypt the received global model and apply the decrypted parameters to update their local models.

(a) Model Decryption at PGO: On reception of the global model broadcasted by the DC, each PGO decrypts the global model $[[w_{\text{global}}^c]]$ into

plaintext using the private key SK with the following equation:

$$\text{Decryption}\left(\text{SK}, \left[\left[w_{\text{global}}^c\right]\right]\right) \rightarrow w_{\text{global}}^c \quad (9)$$

(b) Local Update at PGO: The global model is loaded into the local model of the PGO, and repeating stage 2–4 until the model converges or reaches the maximum number of rounds. This step ensures that each local node can use the FL results to improve its own model without sharing sensitive data.

Data set

For data-driven risk assessment, a substantial amount of data is required. However, there is often a scarcity of system risk samples in the real world. Therefore, it is necessary to generate sufficient training data based on historical data. The dataset generation is implemented using MATLAB 2022a.

(1) Data Generation: We first obtain a set of system operating status data by calling the IEEE 14-bus system and the IEEE 118-bus system built into the MATPOWER 7.1 toolbox²⁹. Since the data values of baseKV, zone, conductance G_s , susceptance B_s , maximum bus amplitude $V_{\max}$, and minimum amplitude $V_{\min}$ are fixed in the system, the remaining data types used to characterize the operating state of the power system, such as bus voltages, power injections, and branch power flows, are typically constrained by power flow equations. When the active demand P_d and reactive demand Q_d of a node are changed, running the power flow calculation can yield a different set of system state data. Therefore, referring to the approach in^{10,30}, we extend real world data based on the standard node loads of standard node system. The generated data obeys the normal distribution $X \sim \mathcal{N}(\mu, \sigma^2)$, where the mean μ is equal to the standard P_d or Q_d , and the standard deviation σ is 1/6 of the standard value.

(2) Data Preprocessing: In the IEEE 14-bus system and IEEE 118-bus system, we generated 160,000 data instances each, where each instance consists of the bus active power injection and voltage magnitude level, as well as the power flow loss on branches. First, we applied min-max normalization to the three types of data separately, scaling all data values to the same range. Next, based on the comprehensive risk assessment index system previously constructed, we calculated the comprehensive index risk values corresponding to each instance. The comprehensive risk value reflects the overall security status of the system, providing a comprehensive assessment of various risk indicators. When the comprehensive risk value falls within a small range, the system is in a safe operational state; otherwise, it indicates potential security risks in the system. We divide the risk values into four ranges to represent four risk levels, and then label each instance with its

corresponding risk level. Finally, we split the complete dataset into two parts: the first 80% for training and the remaining 20% for testing.

Experiment settings

(1) Environmental Setup: All experiments are conducted on a cloud server equipped with an AMD EPYC 7642 48-Core Processor, RTX 3090 (24GB) GPU, and 80-GB RAM. The entire FedRA system is developed in a Python 3.8 environment based on PyTorch 1.10.0. For the encryption scheme, we use the Microsoft SEAL open source project CKKS fully homomorphic encryption scheme.

(2) Performance Metrics: To evaluate the performance of the risk assessment scheme, we measure the classification accuracy and loss value of the model. We also selected Precision, Recall and F1-score to evaluate the classification performance of the assessment model for different risk levels. For the multi-classification problem, we can understand the model as an integration of four binary classification models. For the j -th risk level label, we count TP_j (True positive), FP_j (False positive), FN_j (False negative), and TN_j (True negative). Based on the above statistical terms, we can calculate the three metrics for category j using the following equation.

$$\text{Precision}_j = \frac{TP_j}{TP_j + FP_j} \quad (10)$$

$$\text{Recall}_j = \frac{TP_j}{TP_j + FN_j} \quad (11)$$

$$F1_j = \frac{TP_j}{TP_j + FN_j} \quad (12)$$

Then, we calculate the average metrics for all categories by averaging $\{\text{Precision}_j\}_{j=1}^4$, $\{\text{Recall}_j\}_{j=1}^4$, and $\{F1_j\}_{j=1}^4$ separately.

Results

Comparison with other classification models

Experiment 1. This experiment provides a performance comparison of our CNN-based model with three commonly used classification models, i.e., Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN). In this research, a random search strategy is used to tune the hyperparameter combinations of each model. The specific hyperparameter combinations of our CNN-based model are given in Table 1, in which the Adam optimizer is employed with the learning rate η set to 0.001, and the cross-entropy is the loss function.

Table 2 presents the performance of our CNN-based model and other models in terms of accuracy, precision, recall and F1-score under different bus systems. The following conclusions can be drawn:

- Among all models, KNN exhibits the worst performance. This is because for high-dimensional data, the effectiveness of distance metrics deteriorates, leading to distances between data points becoming similar. Consequently, KNN is unable to effectively classify the data.
- While the performance of RF and SVM is good, it remains inferior to that of the CNN-based model. This is due to the convolutional operations in CNN, which process the input globally and capture key risk features in the state data through multiple layers of feature

Table 1 | Hyperparameter settings

Parameters	Value
Training epoch	100
Batch size	256
Optimizer	Adam
Learning rate	0.001
Loss function	Cross-entropy

Table 2 | Comparisons with other models on two bus systems

Metrics	IEEE 14-bus system				IEEE 118-bus system			
	KNN	RF	SVM	CNN	KNN	RF	SVM	CNN
Precision	0.6250	0.7798	0.8450	0.8768	0.7675	0.8353	0.8875	0.8955
Recall	0.5675	0.7746	0.8475	0.8754	0.7350	0.8346	0.8912	0.8947
F1-score	0.5350	0.7751	0.8425	0.8749	0.7325	0.8325	0.8905	0.8948
Accuracy	0.5701	0.7779	0.8462	0.8763	0.7381	0.8305	0.8898	0.8953

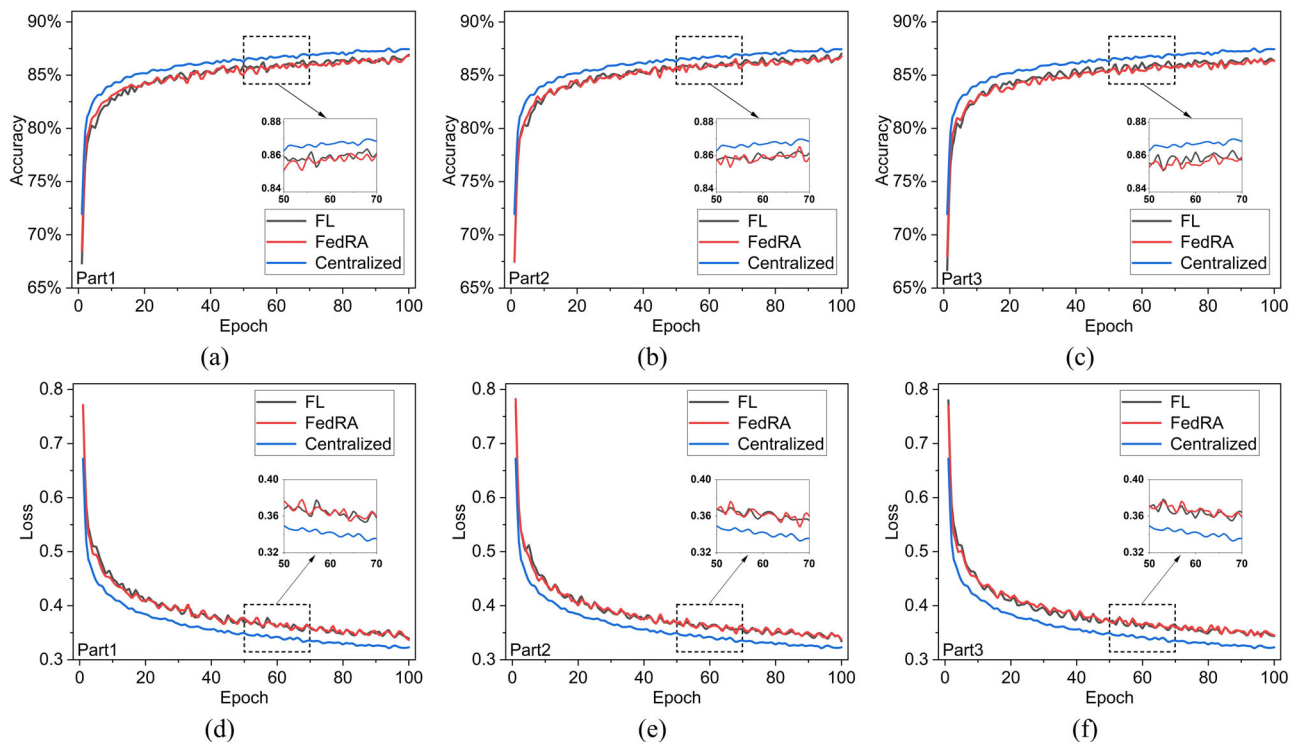


Fig. 4 | Comparison of the performance of different models for three parts on IEEE 14-bus system. Each part is associated with two subplots: one illustrating the alteration in accuracy and the other depicting the modification in loss values. FL: In this scenario, each node only perform the basic federated training, where model parameters are transmitted and aggregated in plaintext without additional privacy measures. Centralized: By gathering all node datasets and training a centralized

model, this method completely disregards the privacy of individual node data. In contrast to local training, it can provide the best model under ideal conditions. FedRA: The method proposed in this paper. **a, d** Model accuracy and loss variation for part 1. **b, e** Model accuracy and loss variation for part 2. **c, f** Model accuracy and loss variation for part 3.

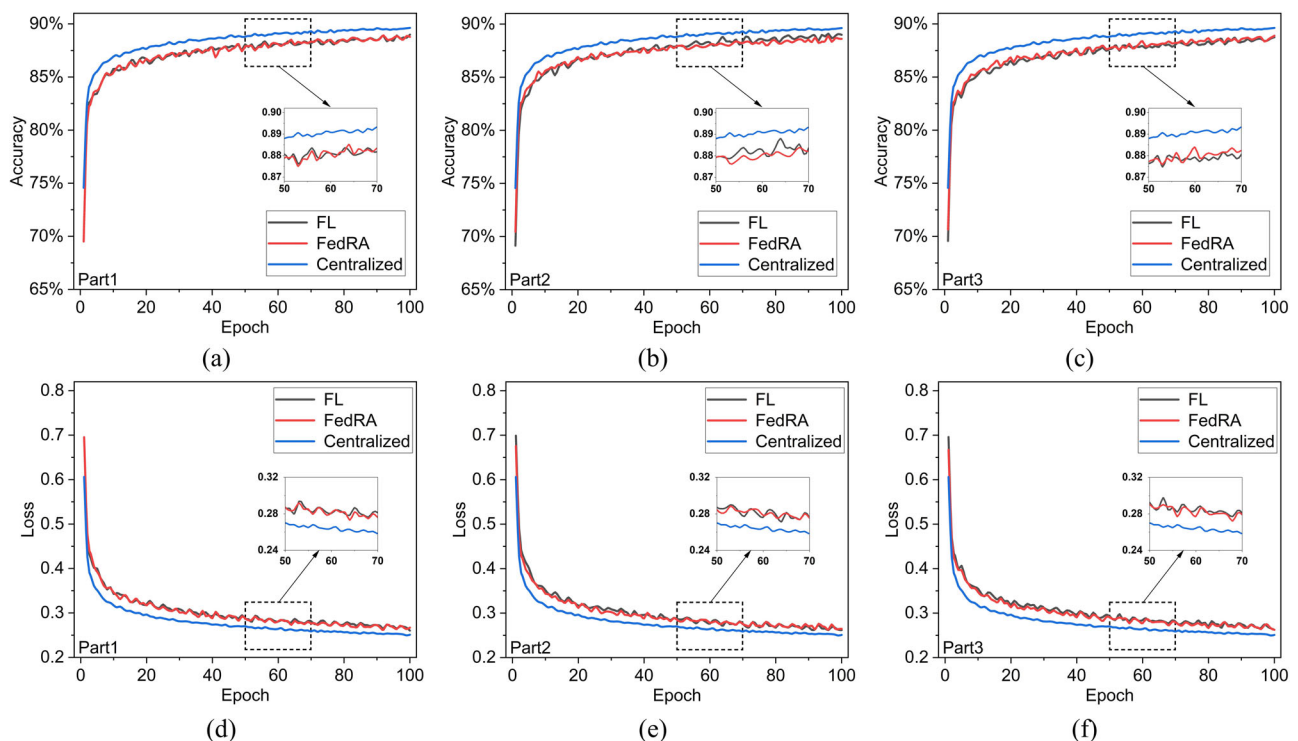
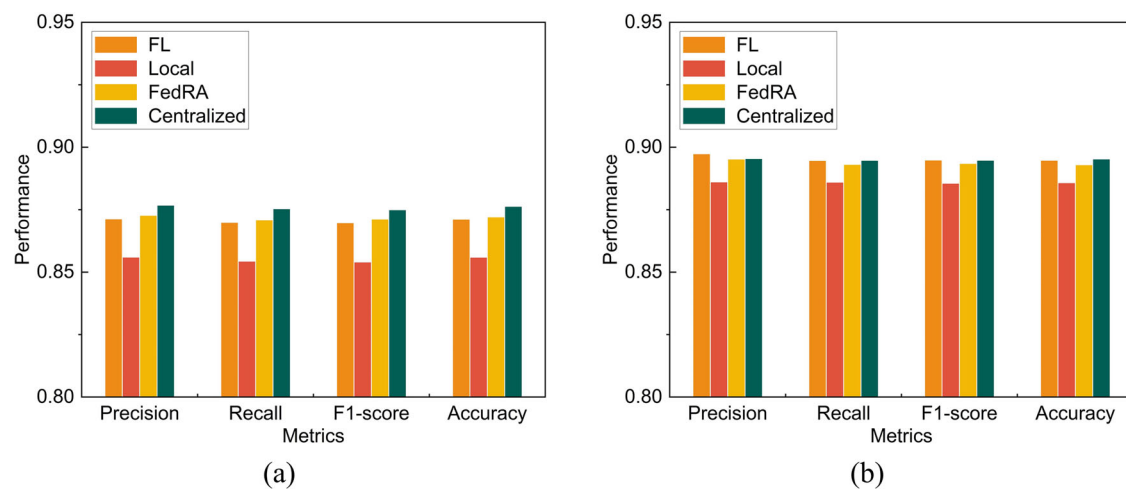


Fig. 5 | Comparison of the performance of different models for three parts on IEEE 118-bus system. **a, d** Model accuracy and loss variation for part 1. **b, e** Model accuracy and loss variation for part 2. **c, f** Model accuracy and loss variation for part 3.

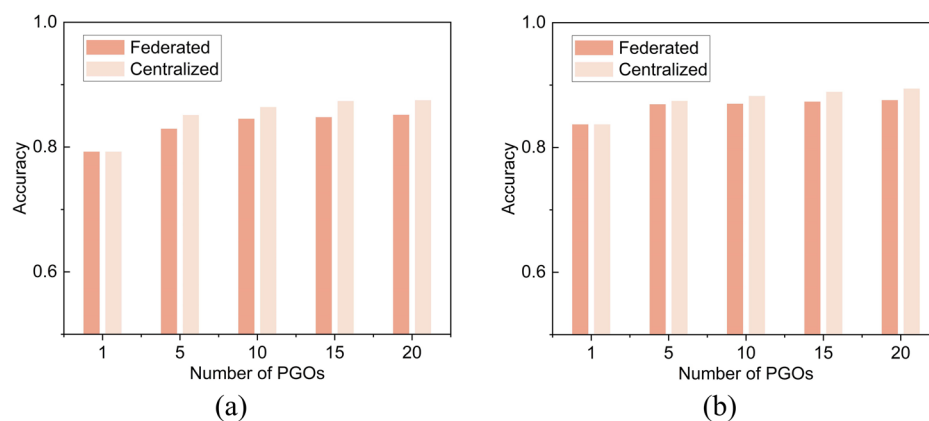
Table 3 | Comparison of recognition performance of different risk levels on IEEE 14-bus system and IEEE 118-bus system

Dataset	Risk level	Centralized			FL			FedRA (our method)		
		Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
IEEE 14-bus system	Safe	0.8642	0.9664	0.9124	0.8646	0.9527	0.9065	0.8721	0.9451	0.9071
	Low	0.8753	0.8151	0.8441	0.8680	0.8193	0.8430	0.8579	0.8298	0.8436
	Medium	0.8645	0.8201	0.8417	0.8528	0.8123	0.8321	0.8376	0.8333	0.8354
	High	0.9034	0.8999	0.9016	0.8999	0.8954	0.8977	0.9234	0.8755	0.8988
IEEE 118-bus system	Safe	0.9300	0.9628	0.9461	0.9204	0.9660	0.9427	0.9287	0.9585	0.9434
	Low	0.9195	0.8639	0.8909	0.9195	0.8607	0.8892	0.9174	0.8547	0.8849
	Medium	0.8394	0.8569	0.8481	0.8193	0.8949	0.8554	0.8244	0.8818	0.8521
	High	0.8930	0.8955	0.8943	0.9304	0.8573	0.8923	0.9105	0.8776	0.8938

**Fig. 6 | Comparison of overall performance of different models on two bus systems.** Local: Considering the privacy constraints of data interaction between multiple nodes, this method involves local training using data from a single node without communication with other nodes. Due to the smaller amount of training

data, this method typically provides the poorest evaluation performance.

a Experimental results on the IEEE 14-bus system. **b** Experimental results on the IEEE 118-bus system.

**Fig. 7 | Accuracy of different models versus the number of nodes on two bus systems.** **a** The model accuracy changes with the number of participating nodes on the IEEE 14-bus system. **b** The model accuracy changes with the number of participating nodes on the IEEE 118-bus system.

representation, thereby enabling effective classification and assessment.

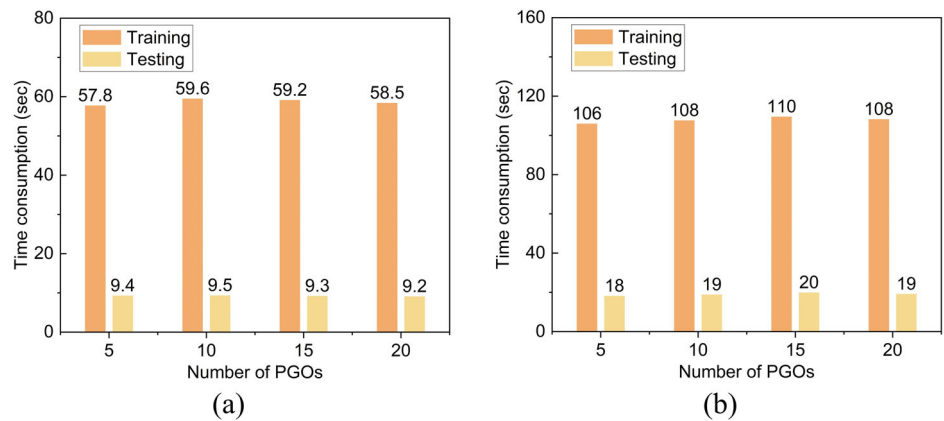
Performance evaluation of each node under different training schemes

Experiment 2. This experiment provides a comprehensive evaluation of the datasets generated in the IEEE 14-bus system and the IEEE 118-bus

system by comparing with three other methods to demonstrate the effectiveness of the proposed FedRA scheme. For the $K = 3$ nodes participating in FL, the data is divided into three parts in a uniform manner, with each PGO node evenly assigned training and testing datasets. Furthermore, the number of local iterations required to communicate with the DC is set to 4. All training methods use the proposed network model as the base model, with the same structure and parameter configuration.

Fig. 8 | Average time consumption per node versus the number of nodes on two bus systems.

a Comparison of average time consumption with different number of nodes on the IEEE 14-bus system. **b** Comparison of average time consumption with different number of nodes on the IEEE 118-bus system.



Figures 4–5 depict the accuracy and loss curves for three nodes using different training schemes on two datasets. By examining these figures, we can draw the following conclusions:

- Firstly, all the models can converge quickly in a few rounds of local training with different users, which shows that the proposed assessment model exhibits excellent convergence performance.
- Our proposed FedRA scheme can provide a good assessment model for all participating users. As the number of iterations increases, the performance of the models gradually improves, ultimately achieving an accuracy of over 85% in both the IEEE 14-bus system and IEEE 118-bus system.
- We observe that both FedRA and FL exhibit slightly lower performance compared to the Centralized approach. This is because the centralized scheme does not consider data privacy and only aims to achieve the best model performance.
- On the other hand, FL's distributed training involves data exchanges among multiple nodes without leaving their local domain. Compared to FL without privacy protection mechanisms, FedRA shows nearly identical performance, suggesting that FedRA maintains model accuracy while ensuring the privacy security of model parameters.

Table 3 shows the performance of different schemes in terms of Precision, Recall, and F1-score for four risk levels in the IEEE 14-bus system and IEEE 118-bus system. Figure 6 further shows the overall performance differences of the models, and we also tested the performance of local training models to validate the effectiveness of our scheme. The following conclusions can be drawn:

- From Table 3, it can be observed that all models achieved high scores in the recognition of risk level Safe, and risk level High Risk. This is because for power system state data, the risk level is directly proportional to the degree of data damage: a lower risk level means smaller differences between damaged and normal data, while a higher risk level means substantial differences. Therefore, risk assessment models can more easily identify and analyze these two risk levels from the data.
- In the overall performance test experiments, the local model provided the expected worst performance, while our scheme clearly outperforms it. Compared to the ideal model obtained from centralized training, FL does not always perform worse than the centralized approach. In particular, FedRA demonstrates superior performance over FL on the IEEE 14-bus system, indicating that it can ensure model performance while protecting the privacy of data at each node.
- Due to the larger amount of state data in the IEEE 118-bus system, the increased size of parameters in the risk assessment model led to more parameter losses after homomorphic computations. Overall, FedRA still achieves comparable performance to the baseline models. Furthermore, our goal is not to outperform the other two baseline models. Considering the substantial improvement in maintaining the

privacy value of node data, we find that a smaller loss in model performance is acceptable.

Scalability performance with multiple nodes

Experiment 3. This experiment studies the scalability and applicability of our FedRA scheme with more nodes, in which we first compare the model accuracy across different numbers of nodes, and then summarize the model's time efficiency. For this purpose, we divided the entire dataset equally into 20 parts to simulate a scenario with up to $K = 20$ PGOs. Additionally, an ideal model trained using data from the centralized K nodes was used as the baseline.

Figure 7 shows the variations in global model accuracy for this research's method and the centralized method under different bus systems, as the number of nodes participating in training changes. The following conclusions can be drawn:

- As the number of nodes increases, the performance of both methods is improved. This is because the amount of data involved in training increases, thus enhancing the training effectiveness. Meanwhile, the performance of FedRA is almost as well as the centralized method, but the former takes data privacy into account.
- Especially for $K = 1$, it also represents the performance of local training on a single node. Despite the fixed amount of data for a single node, as the number of participating nodes increases, each PGO node benefits from federated learning, thereby achieving enhanced performance compared to the local model.

Figure 8 shows the average time consumption for each node with varying numbers of nodes during the entire federated process using this research's method under different bus systems. It can be observed that the average time efficiency of the nodes remains almost constant as the number of nodes increases. In contrast, for more complex 118-bus systems, the average time consumption increases, but the greater data features also result in greater accuracy gains.

Conclusions

To collaborate with power nodes for distributed risk assessment while protecting data privacy, we propose a federated risk assessment method that considers privacy protection. We begin by analyzing the impact of risk events and constructing a comprehensive risk assessment indicator system, which we use to generate an extended dataset. Next, we train a local model in a data-driven manner using a CNN-based risk assessment model. Finally, a secure federated assessment scheme was designed to collaborate with the nodes to train a global assessment model. The homomorphic encryption scheme was employed to ensure secure aggregation and communication of model parameters. Comparative experimental results on the IEEE 14-bus system and the IEEE 118-bus system indicate that 1) the FedRA scheme provides good global models for all involved users, and our models have

good accuracy and convergence speed; 2) compared to the ideal model under centralized training, although FedRA loses some of the evaluation accuracy while providing privacy protection, we consider that the value of privacy protection is much higher than the value of small model performance improvement; and 3) compared to local models and FL models, FedRA can both safely aggregate model parameters and achieve high accuracy, especially in identifying high-risk states of a system. Our future plan is to combine multiple data sources to provide a more comprehensive and multidimensional assessment of grid risk.

Data availability

The detailed data sets are available at GitHub³¹.

Code availability

The source code is available at GitHub³¹.

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Author contributions

Song Deng and Longxiang Zhang conceived the experiments, Longxiang Zhang conducted the experiments, Song Deng and Longxiang Zhang analysed the results. All authors reviewed the manuscript. Song Deng and Dong Yue provide the funding support.

Competing interests

The authors declare no competing interests.

Additional information

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